

SALALAH, Oman

WMO: 413160

Lat: 17.039N

Lon: 54.091E

Elev: 73

StdP: 14.66

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	64.3	66.1	19.1	14.4	74.8	24.8	18.9	77.3	26.1	72.2	23.2	73.9	7.0	350	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.0	92.8	71.0	91.2	74.9	89.8	76.1	82.5	87.1	81.7	86.6	81.0	86.3	12.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.2	86.0	80.5	158.7	85.7	79.2	151.9	84.9	46.6	88.3	45.8	86.8	45.0	86.7	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.9	15.9	60.3	102.4	2.0	5.4	58.9	106.2	57.8	109.4	56.7	112.5	55.3	116.4		
			44.3	83.2	2.3	1.6	42.6	84.4	41.2	85.3	39.9	86.2	38.3	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	74.0	75.2	78.9	82.8	85.7	85.3	79.8	77.9	80.5	80.8	79.9	76.9
	DBStd	4.20	2.42	2.97	2.87	2.32	1.90	2.18	2.49	1.90	1.82	2.32	2.45	2.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10885	743	705	895	983	1107	1059	923	866	915	956	898	835
	CDD65	5410	278	285	430	533	642	609	458	401	465	491	448	370
	CDH74	54166	1553	1949	4094	6444	8590	7801	3997	2636	4380	5286	4563	2873
	CDH80	17379	134	295	1112	2538	4176	3508	655	174	887	1822	1484	594
(14) Wind	WSAvg	7.1	7.6	7.0	6.4	6.7	7.8	8.7	7.2	6.3	7.0	5.7	6.2	8.4
(15) Precipitation	PrecAvg	2.9	0.0	0.0	0.2	0.4	0.1	0.1	0.9	0.9	0.1	0.2	0.1	0.0
	PrecMax	6.8	0.4	0.4	3.6	5.9	1.7	0.6	2.4	2.5	0.6	2.7	1.7	0.5
	PrecMin	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.3	0.1	0.1	0.6	1.1	0.3	0.1	0.5	0.5	0.2	0.5	0.3	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.4	89.4	93.9	97.0	94.3	92.8	88.0	84.6	87.7	94.9	93.6	89.7
		MCWB	60.4	61.5	62.8	64.7	73.6	81.5	80.1	77.5	77.1	67.1	64.8	62.8
	2%	DB	82.8	85.8	89.8	91.1	91.6	91.2	86.1	82.8	86.2	90.3	89.9	86.4
		MCWB	61.8	61.3	63.5	71.1	79.1	80.9	79.4	76.8	76.3	69.5	64.9	63.1
	5%	DB	81.0	82.8	87.3	89.3	90.5	89.8	84.4	82.1	85.2	88.1	87.8	84.5
		MCWB	64.6	65.2	66.7	74.7	79.1	80.4	78.6	76.6	75.9	72.3	67.7	64.6
	10%	DB	80.3	81.0	84.6	87.8	89.7	89.2	83.1	80.7	84.3	86.4	86.1	82.8
		MCWB	65.2	66.9	70.1	75.6	79.0	80.3	78.0	76.3	75.6	73.6	70.0	66.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.6	74.6	77.4	81.0	83.1	84.1	81.5	79.2	79.4	78.9	77.9	75.1
		MCDB	79.1	80.4	82.9	86.5	89.3	88.8	84.7	81.9	83.2	84.4	84.0	81.2
	2%	WB	71.2	72.9	76.2	79.6	81.8	83.0	80.3	78.4	78.7	77.9	76.4	72.9
		MCDB	78.3	79.7	82.5	86.0	87.6	87.5	83.6	81.0	82.4	84.0	83.6	80.3
	5%	WB	70.0	71.9	75.3	78.8	81.0	82.3	79.5	77.8	78.1	77.1	75.2	71.8
		MCDB	77.6	79.2	82.1	85.5	87.3	86.7	82.8	80.3	82.0	83.7	83.0	80.0
	10%	WB	69.0	70.7	74.4	78.0	80.4	81.6	78.8	77.1	77.4	76.4	74.2	70.7
		MCDB	77.2	78.4	81.5	85.0	87.0	86.4	82.0	79.5	81.5	83.3	82.4	79.6
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	13.5	13.1	11.2	9.0	6.7	5.0	4.9	7.4	13.0	14.2	13.6
		MCDBR	14.4	15.4	18.0	13.6	9.6	7.8	6.4	6.4	8.5	15.9	16.5	14.6
		MCWBR	9.5	11.1	13.7	9.9	4.9	3.6	2.6	2.6	4.0	10.9	11.3	9.3
	5% WB	MCDBR	12.0	11.8	10.5	10.0	8.8	7.0	5.7	5.5	7.0	11.1	12.9	12.5
		MCWBR	8.3	8.4	7.4	6.3	4.7	3.6	2.7	2.7	3.6	8.1	9.6	8.7
(40) Clear-Sky Solar Irradiance	taub		0.420	0.443	0.495	0.527	0.618	0.791	0.834	0.769	0.620	0.459	0.438	0.421
	taud		2.180	2.101	1.962	1.921	1.738	1.461	1.428	1.528	1.756	2.117	2.161	2.176
	Ebn at Noon		273	272	262	253	228	190	182	196	227	264	265	268
	Edh at Noon		45	51	60	63	74	97	101	92	73	49	45	44
(44) All-Sky Solar Radiation	RadAvg		1728	1984	2205	2334	2339	2096	1799	1825	1994	2034	1775	1655
	RadStd		46	50	78	63	79	103	95	60	81	51	62	53

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.27	N/A	-2.74	N/A	N/A	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page